

File permissions in Linux

Project description

For this activity, I worked with file and directory permissions in the `projects` directory. The main goal was to check the permissions that were already set and then change them so they matched the access requirements given by the organization.

Check file and directory details

I started by checking the contents and permissions of the `projects` directory. I used the following command:

`ls -la`

```
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w--- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx--x-- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

I used the `-l` option to get detailed information, including file permissions, and the `-a` option so that hidden files would also be displayed.

From the output, I could see a directory named `drafts`, a hidden file named `.project_x.txt`, and several other project files. The first column in the output showed a 10-character permissions string for each item.

Describe the permissions string

The permissions string tells me what type of item it is and what access the user, group, and other users have.

For example, the permissions for `project_t.txt` were:

`-rw-rw-r--`

The first character is `-`, which means it is a regular file. If this character were `d`, it would mean the item was a directory.

The remaining nine characters are divided into three groups:

- The first three characters show the permissions for the **user/owner**.
- The next three show the permissions for the **group**.
- The last three show the permissions for **other users**.

The letters have the following meanings:

- `r` = read
- `w` = write
- `x` = execute
- `-` = permission is not granted

So, in `-rw-rw-r--`, the user and group have read and write permissions, while other users only have read permission. Nobody has execute permission.

Change file permissions

The organization required that other users should not have write access to any of the project files. After checking the existing permissions, I identified `project_k.txt` as a file that needed to be changed.

```
researcher2@5d738f0f927b:~/projects$ chmod o-w project_k.txt
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w--- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

I used the `chmod` command to remove write permission from other users. After making the change, I ran `ls -la` again to confirm that the permissions had been updated correctly.

The `chmod` command is useful for modifying permissions, and the permission specification tells Linux which permission should be changed and for which category of users.

Change file permissions on a hidden file

Next, I worked with the hidden file `.project_x.txt`. Since its name begins with a period (`.`), Linux treats it as a hidden file.

```
researcher2@3213bbc1d047:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3213bbc1d047:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 ..
-r--r----- 1 researcher2 research_team  46 Dec 20 15:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec 20 15:36 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Dec 20 15:36 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec 20 15:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec 20 15:36 project_t.txt
researcher2@3213bbc1d047:~/projects$
```

The requirement was that nobody should have write access to this file, while the user and group should have read access.

I used `chmod` to remove write permission from the user and group and to make sure the group had read permission. In particular:

- `u-w` removed write permission from the user.
- `g-w` removed write permission from the group.
- `g+r` added read permission for the group.

I then checked the permissions again to make sure the changes were applied as intended.

Change directory permissions

Finally, I updated the permissions for the `drafts` directory. The requirement was that only `researcher2` should have execute access to this directory and its contents.

```
researcher2@5d738f0f927b:~/projects$ chmod g-x drafts
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-r--r----- 1 researcher2 research_team  46 Dec  2 15:27 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

I checked the existing permissions and found that the group had executed permission. Since the group should not have this access, I used `chmod` to remove the group's execute permission.

The `researcher2` user already had execute permission, so I did not need to add it again.

After making the change, I used the permission listing to verify that the directory had the required permissions.

Summary

In this activity, I reviewed and modified several Linux file and directory permissions in the `projects` directory. I first used `ls -la` to see the existing permissions, including those of hidden files. I then used `chmod` to make the required changes to `project_k.txt`, `.project_x.txt`, and the `drafts` directory.

This helped me understand how Linux permissions work and how the `u`, `g`, and `o` categories can be used with `chmod` to control access for users, groups, and other users.